students) whether the star that formed the hole was made of matter or antimatter, of protons and electrons, or of neutrinos and antineutrinos. In Wheeler's words, made more precise, a black hole has *almost* no hair; its only "hair" is its mass, its spin, and its electric charge.

The firm, ultimate proof that a black hole has no hair (except its mass, spin, and electric charge) was actually not Price's. Price's analysis was restricted to imploding stars that are very nearly spherical, and that spin, if at all, only very slowly. The perturbation methods he used required this restriction. To learn the ultimate fate of a highly deformed, rapidly spinning, imploding star required a set of mathematical techniques very different from perturbation methods.

Dennis Sciama's students at Cambridge University were masters of the required techniques, but the techniques were difficult; extremely so. It took fifteen years for Sciama's students and their intellectual descendants, using those techniques, to produce a firm and complete proof that black holes have no hair—that even if a hole spins fast and is strongly deformed by its spin, the hole's final properties (after all radiation has flown away) are uniquely fixed by the hole's mass, spin, and charge. The lion's share of the credit for the proof goes to two of Sciama's students, Brandon Carter and Stephen Hawking, and to Werner Israel; but major contributions came also from David Robinson, Gary Bunting, and Pavel Mazur.

In Chapter 3, I commented on the great difference between the laws of physics in our real Universe and the society of ants in T. H. White's epic novel *The Once and Future King*. White's ants were governed by the motto "Everything not forbidden is compulsory," but the laws of physics violate that motto flagrantly. Many things allowed by physical law are so highly improbable that they never occur. Price's theorem is a remarkable exception. It is one of the few situations I have ever encountered in physics where the ants' motto holds sway: If physical law does not forbid a black hole to eject something as radiation, then ejection is compulsory.

Equally unusual are the implications of a black hole's resulting "hairless" state. Normally we physicists build simplified theoretical or computer models to try to understand the complicated Universe around us. As an aid to understanding weather, atmospheric physicists build computer